

Andrew Mouchantaf

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Skills

Hardware / Embedded: STM32, ESP32, Edge ML, Embedded Linux, RTOS, PCIe, I2C, SPI, UART, Interrupts, DMA

Software: C, C++, CMake, Python, Kotlin, Docker, Kubernetes, Kafka, gRPC, Protobuf, REST, GraphQL, PostgreSQL, CockroachDB, CI/CD pipelines

Mechanical: Fusion 360, SolidWorks, AutoCAD, MATLAB/Simulink

Experience

Embedded Systems Engineer

Mar 2026 - Present

Deep Breathe

Toronto, ON

- Led a cross-functional team to develop a custom ultrasound probe with integrated AI pathology inference, planning the development roadmap and driving the project through funding and regulatory milestones
- Built a custom probe prototype with a 3D-printed shell, an ESP32-S3 MCU, a touch LCD screen, and a docking station tablet interface, demoing the device to ICU clinicians to solicit early design feedback
- Quantized models and designed a benchmark methodology to analyze performance of various candidate processors to enable a compact form factor and minimize manufacturing costs and inference latency
- Designed a device entitlement and identity platform for managing access and data retrieval for organization users
- Optimized model library to leverage new model format, reducing FAST and hemothorax inference latencies by **30%**

Software Engineer

Aug 2022 - Dec 2025

DoorDash

Seattle, WA

- Led new authorization service project with updated profiles and role-based access, migrating all **~2M** DoorDash merchant users with **zero downtime and incidents** through data transformation jobs and request shadowing
- Implemented a graph data model and search API, replacing a legacy tree structure and migrating **~8M** DoorDash vendor entities to reduce support requests and enable **>\$10M** in annual revenue from new product verticals
- Delivered a consolidation platform for Kubernetes cost and utilization across all DoorDash engineering deployments via data-collector pods and ETL pipelines, identifying **~\$10M** annual savings from overprovisioned resources
- Architected an internal cost alerting platform with data pipelines, cost analysis workflows, and a presentation API to detect spending anomalies across DoorDash engineering teams, delivering **~\$500k** in average annual savings
- Developed request hedging tools at the orchestration layer, improving responsiveness for unreliable services by **~70%** and reducing false alerts to on-call engineers

Projects

TCP Connection Analyzer

Jan 2026

- Developed a real-time passive connection analyzer in C using raw Linux packet sockets with multi-layer parsing for Ethernet, IPv4, and TCP headers with checksum validation and flag decoding

Facial Expression Operated Motorized Wheelchair (Capstone Project)

Sep 2021 - Apr 2022

- Created a motorized wheelchair attachment controlled via an EMG headset parsing facial expressions to improve mobility and independence for people with quadriplegia

Education

University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering (Honours with Co-op)

Sep 2017 - Jun 2022

Waterloo, ON